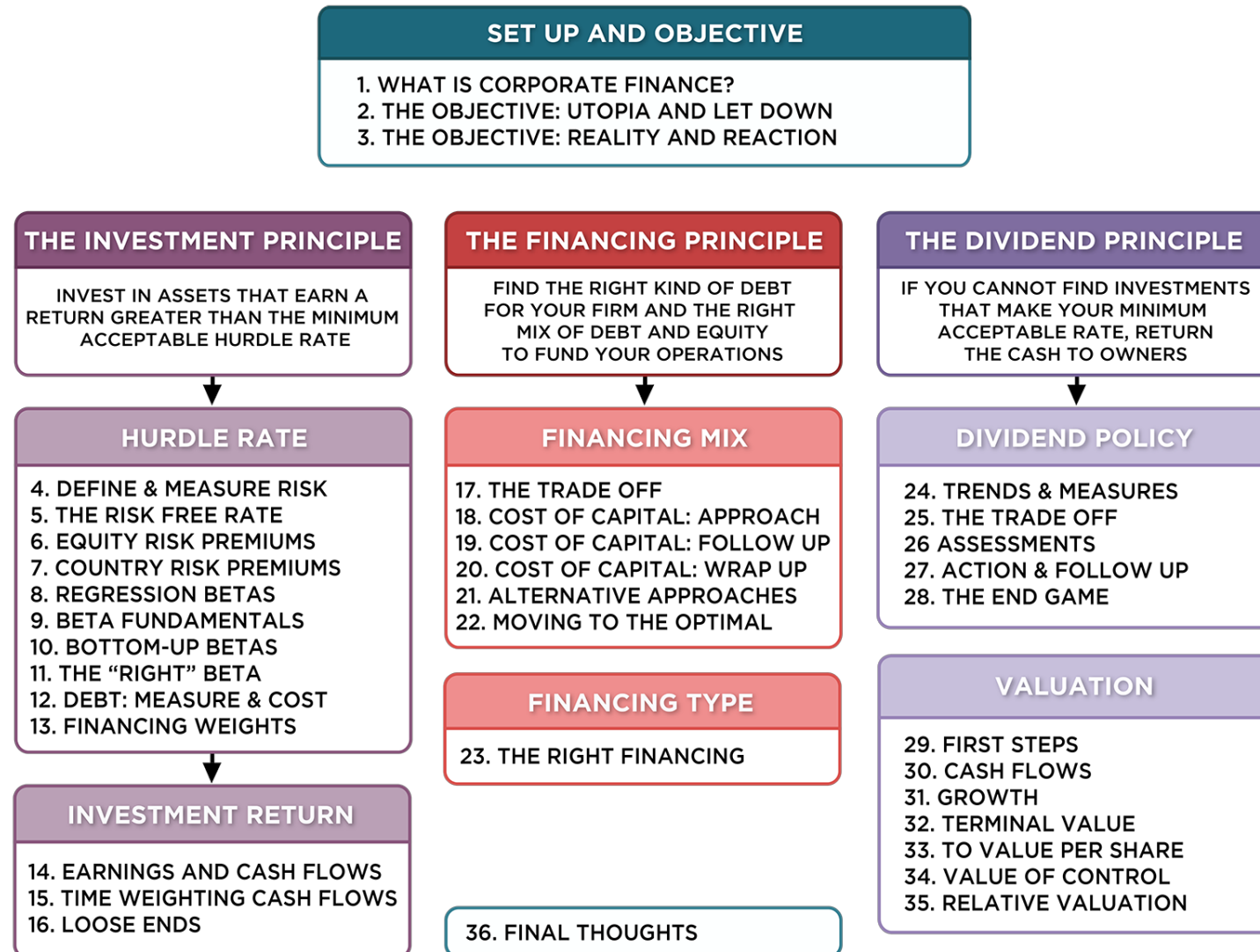


SESSION 20: COST OF CAPITAL: WRAP UP

One size does not fit all.

SESSION 20: COST OF CAPITAL: WRAP UP



ENHANCED COST OF CAPITAL APPROACH

- Distress cost affected operating income: In the enhanced cost of capital approach, the indirect costs of bankruptcy are built into the expected operating income. As the rating of the firm declines, the operating income is adjusted to reflect the loss in operating income that will occur when customers, suppliers and investors react.
- Dynamic analysis: Rather than look at a single number for operating income, you can draw from a distribution of operating income (thus allowing for different outcomes).

ESTIMATING THE DISTRESS EFFECT – DISNEY

<i>Rating</i>	<i>Drop in EBITDA (Low)</i>	<i>Drop in EBITDA (Medium)</i>	<i>Drop in EBITDA (High)</i>
To A	No effect	No effect	2.00%
To A-	No effect	2.00%	5.00%
To BBB	5.00%	10.00%	15.00%
To BB+	10.00%	20.00%	25.00%
To B-	15.00%	25.00%	30.00%
To C	25.00%	40.00%	50.00%
To D	30.00%	50.00%	100.00%

THE OPTIMAL DEBT RATIO WITH INDIRECT BANKRUPTCY COSTS

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	0.9239	8.07%	Aaa/AAA	3.15%	36.10%	2.01%	8.07%	\$122,633
10%	0.9895	8.45%	Aaa/AAA	3.15%	36.10%	2.01%	7.81%	\$134,020
20%	1.0715	8.92%	Aaa/AAA	3.15%	36.10%	2.01%	7.54%	\$147,739
30%	1.1769	9.53%	Aa2/AA	3.45%	36.10%	2.20%	7.33%	\$160,625
40%	1.3175	10.34%	A2/A	3.75%	36.10%	2.40%	7.16%	\$172,933
50%	1.5573	11.72%	C2/C	11.50%	31.44%	7.88%	9.80%	\$35,782
60%	1.9946	14.24%	Caa/CCC	13.25%	22.74%	10.24%	11.84%	\$25,219
70%	2.6594	18.07%	Caa/CCC	13.25%	19.49%	10.67%	12.89%	\$21,886
80%	3.9892	25.73%	Caa/CCC	13.25%	17.05%	10.99%	13.94%	\$19,331
90%	7.9783	48.72%	Caa/CCC	13.25%	15.16%	11.24%	14.99%	\$17,311

The optimal debt ratio stays at 40% but the cliff becomes much steeper.

EXTENSION TO A FAMILY GROUP COMPANY: TATA MOTORS' OPTIMAL CAPITAL STRUCTURE

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	0.8601	12.76%	Aaa/AAA	9.22%	32.45%	6.23%	12.76%	1,286,997₹
10%	0.9247	13.22%	Aa2/AA	9.52%	32.45%	6.43%	12.54%	1,333,263₹
20%	1.0054	13.80%	A3/A-	10.12%	32.45%	6.84%	12.41%	1,363,774₹
30%	1.1092	14.55%	B2/B	15.32%	32.45%	10.35%	13.29%	1,185,172₹
40%	1.2475	15.54%	Caa/CCC	17.57%	32.45%	11.87%	14.07%	1,061,143₹
50%	1.4412	16.93%	Ca2/CC	18.32%	32.45%	12.38%	14.65%	984,693₹
60%	1.7610	19.23%	Ca2/CC	18.32%	30.18%	12.79%	15.37%	904,764₹
70%	2.3749	23.65%	C2/C	19.32%	24.53%	14.58%	17.30%	741,800₹
80%	3.5624	32.19%	C2/C	19.32%	21.46%	15.17%	18.58%	663,028₹
90%	7.1247	57.81%	C2/C	19.32%	19.08%	15.63%	19.85%	599,379₹

Tata Motors looks like it is over levered (29% actual versus 20% optimal), perhaps because it is drawing on the debt capacity of other companies in the Tata Group.

EXTENSION TO A FIRM WITH VOLATILE EARNINGS: VALE'S OPTIMAL DEBT RATIO

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	0.8601	12.76%	Aaa/AAA	9.22%	32.45%	6.23%	12.76%	1,286,997₹
10%	0.9247	13.22%	Aa2/AA	9.52%	32.45%	6.43%	12.54%	1,333,263₹
20%	1.0054	13.80%	A3/A-	10.12%	32.45%	6.84%	12.41%	1,363,774₹
30%	1.1092	14.55%	B2/B	15.32%	32.45%	10.35%	13.29%	1,185,172₹
40%	1.2475	15.54%	Caa/CCC	17.57%	32.45%	11.87%	14.07%	1,061,143₹
50%	1.4412	16.93%	Ca2/CC	18.32%	32.45%	12.38%	14.65%	984,693₹
60%	1.7610	19.23%	Ca2/CC	18.32%	30.18%	12.79%	15.37%	904,764₹
70%	2.3749	23.65%	C2/C	19.32%	24.53%	14.58%	17.30%	741,800₹
80%	3.5624	32.19%	C2/C	19.32%	21.46%	15.17%	18.58%	663,028₹
90%	7.1247	57.81%	C2/C	19.32%	19.08%	15.63%	19.85%	599,379₹

	Last 12 months	-1	-2	-3	Average
Revenues	\$48,469	\$48,058	\$61,123	\$47,343	\$51,248
EBITDA	\$19,861	\$17,662	\$34,183	\$26,299	\$24,501
EBIT	\$15,487	\$13,346	\$30,206	\$23,033	\$20,518
Pre-tax operating margin	31.95%	27.77%	49.42%	48.65%	39.45%

Replacing Vale's current operating income with the average over the last three years pushes up the optimal to 50%.

OPTIMAL DEBT RATIO FOR A YOUNG, GROWTH FIRM: BAIDU

Debt Ratio	Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	1.3021	12.54%	Aaa/AAA	4.70%	25.00%	3.53%	12.54%	\$337,694
10%	1.4106	13.29%	A3/A-	5.60%	25.00%	4.20%	12.38%	\$343,623
20%	1.5463	14.23%	Ca2/CC	13.80%	25.00%	10.35%	13.45%	\$306,548
30%	1.7632	15.74%	Caa/CCC	14.80%	17.38%	12.23%	14.68%	\$272,853
40%	2.0675	17.85%	D2/D	16.30%	11.83%	14.37%	16.46%	\$235,510
50%	2.4810	20.72%	D2/D	16.30%	9.47%	14.76%	17.74%	\$214,337
60%	3.1012	25.02%	D2/D	16.30%	7.89%	15.01%	19.02%	\$196,657
70%	4.1350	32.20%	D2/D	16.30%	6.76%	15.20%	20.30%	\$181,672
80%	6.2024	46.54%	D2/D	16.30%	5.92%	15.34%	21.58%	\$168,808
90%	12.4049	89.59%	D2/D	16.30%	5.26%	15.44%	22.86%	\$157,646

The optimal debt ratio for Baidu is between 0 and 10%, close to its current debt ratio of 5.23%, and much lower than the optimal debt ratios computed for Disney, Vale and Tata Motors.

EXTENSION TO A PRIVATE BUSINESS: OPTIMAL DEBT RATIO FOR BOOKSCAPE

- Debt value of leases = \$12.136 million (only debt)
- Estimated market value of equity = Net Income * Average PE for Publicly Traded Book Retailers = $1.575 * 20 = \$31.5$ million
- Debt ratio = $12,136 / (12,136 + 31,500) = 27.81\%$

Debt Ratio	Total Beta	Cost of Equity	Bond Rating	Interest rate on debt	Tax Rate	Cost of Debt (after-tax)	WACC	Enterprise Value
0%	1.3632	10.25%	Aaa/AAA	3.15%	40.00%	1.89%	10.25%	\$37,387
10%	1.4540	10.75%	Aaa/AAA	3.15%	40.00%	1.89%	9.86%	\$39,416
20%	1.5676	11.37%	A1/A+	3.60%	40.00%	2.16%	9.53%	\$41,345
30%	1.7137	12.18%	A3/A-	4.05%	40.00%	2.43%	9.25%	\$43,112
40%	1.9084	13.25%	Caa/CCC	11.50%	40.00%	6.90%	10.71%	\$35,224
50%	2.2089	14.90%	Ca2/CC	12.25%	37.96%	7.60%	11.25%	\$32,979
60%	2.8099	18.20%	C2/C	13.25%	29.25%	9.37%	12.91%	\$27,598
70%	3.7466	23.36%	C2/C	13.25%	25.07%	9.93%	13.96%	\$25,012
80%	5.6198	33.66%	C2/C	13.25%	21.93%	10.34%	15.01%	\$22,869
90%	11.4829	65.91%	D2/D	14.75%	17.51%	12.17%	17.54%	\$18,952

The firm value is maximized (and the cost of capital is minimized) at a debt ratio of 30%. At its existing debt ratio of 27.81%, Bookscape is at its optimal.

CAPITAL STRUCTURE FOR A BANK: AN ALTERNATIVE APPROACH

- Consider a bank with \$100 million in outstanding loans and a book value of equity of \$6 million. Furthermore, assume that the regulatory requirement is that equity capital be maintained at 5% of loans outstanding. Finally, assume that this bank wants to increase its loan base by \$50 million to \$150 million and to augment its equity capital ratio to 7% of loans outstanding.

Loans outstanding after expansion	= \$150 million
Equity after expansion	= 7% of \$15 = \$10.5 million
Existing equity	= \$6 million
New equity needed	= \$4.5 million

- Your need for “external” equity as a bank/financial service company will depend upon
 - Your growth rate: Higher growth → More external equity
 - Existing capitalization vs Target capitalization: Under capitalized → More external equity
 - Current earnings: Less earnings → More external equity
 - Current dividends: More dividends → More external equity

DETERMINANTS OF THE OPTIMAL DEBT RATIO:

1. MARGIN TAX RATE

- The primary benefit of debt is tax benefit. The higher the marginal tax rate, the greater the benefit to borrowing:

Tax Rate	Disney	Vale	Tata Motors	Baidu	Bookscape
0%	0%	0%	0%	0%	0%
10%	20%	0%	0%	0%	10%
20%	40%	0%	10%	10%	30%
30%	40%	30%	20%	10%	30%
40%	40%	40%	20%	10%	30%
50%	40%	40%	20%	10%	30%

2. PRE-TAX CASH FLOW RETURN

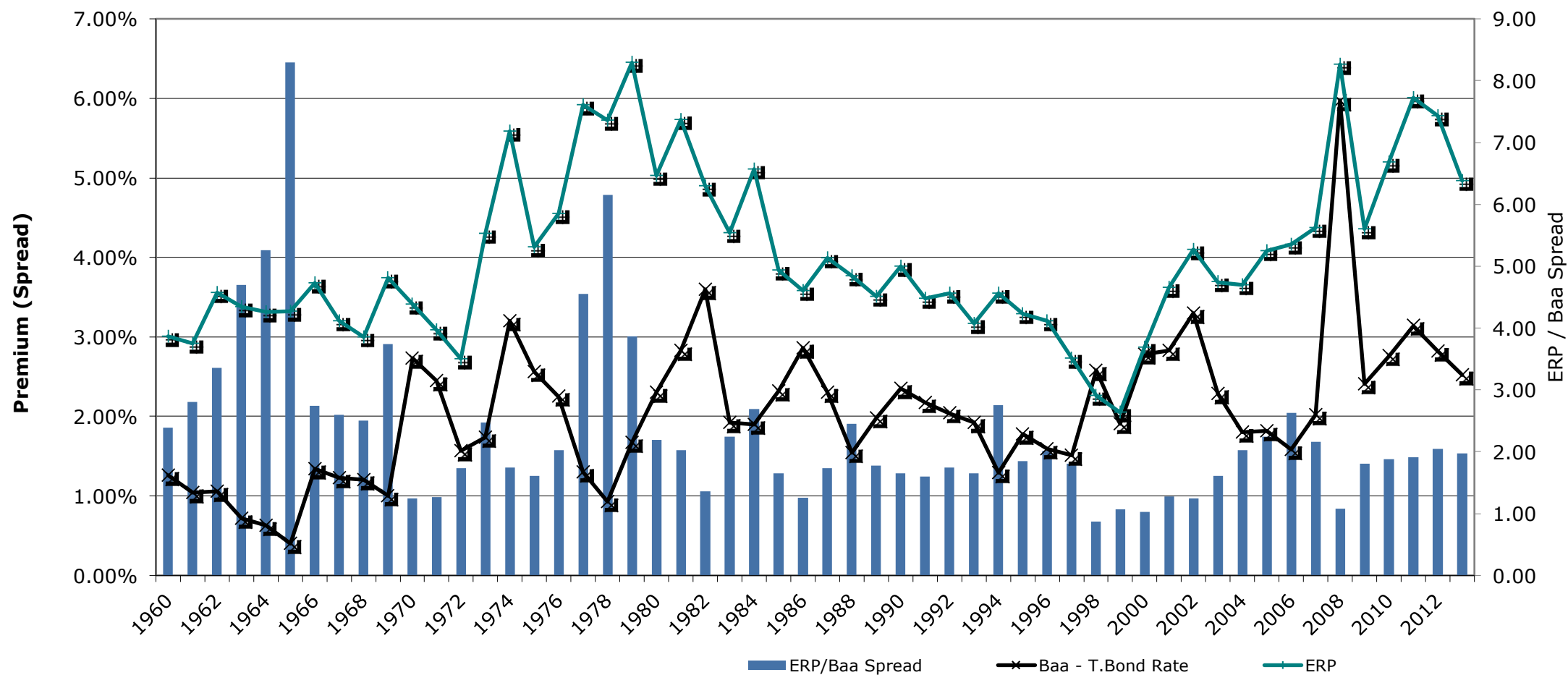
Company	EBITDA	EBIT	Enterprise Value	EBITDA/EV	EBIT/EV	Optimal Debt	Optimal Debt Ratio
Disney	\$12,517	\$10,032	\$133,908	9.35%	7.49%	\$55,136	40.00%
Vale	\$20,167	\$15,667	\$112,352	17.95%	13.94%	\$35,845	30.00%
Tata Motors	250,116₹	166,605₹	1,427,478₹	17.52%	11.67%	325,986₹	20.00%
Baidu	¥13,073	¥10,887	¥342,269	3.82%	3.18%	¥35,280	10.00%
Bookscape	\$4,150	\$2,536	\$42,636	9.73%	5.95%	\$13,091	30.00%

3. OPERATING RISK

- Firms that face more risk or uncertainty in their operations (and more variable operating income as a consequence) will have lower optimal debt ratios than firms that have more predictable operations.
- Operating risk enters the cost of capital approach in two places:
 - Unlevered beta: Firms that face more operating risk will tend to have higher unlevered betas. As they borrow, debt will magnify this already large risk and push up costs of equity much more steeply.
 - Bond ratings: For any given level of operating income, firms that face more risk in operations will have lower ratings. The ratings are based upon normalized income.

4. THE ONLY MACRO DETERMINANT: EQUITY VS DEBT RISK PREMIUMS

Figure 16: Equity Risk Premiums and Bond Default Spreads

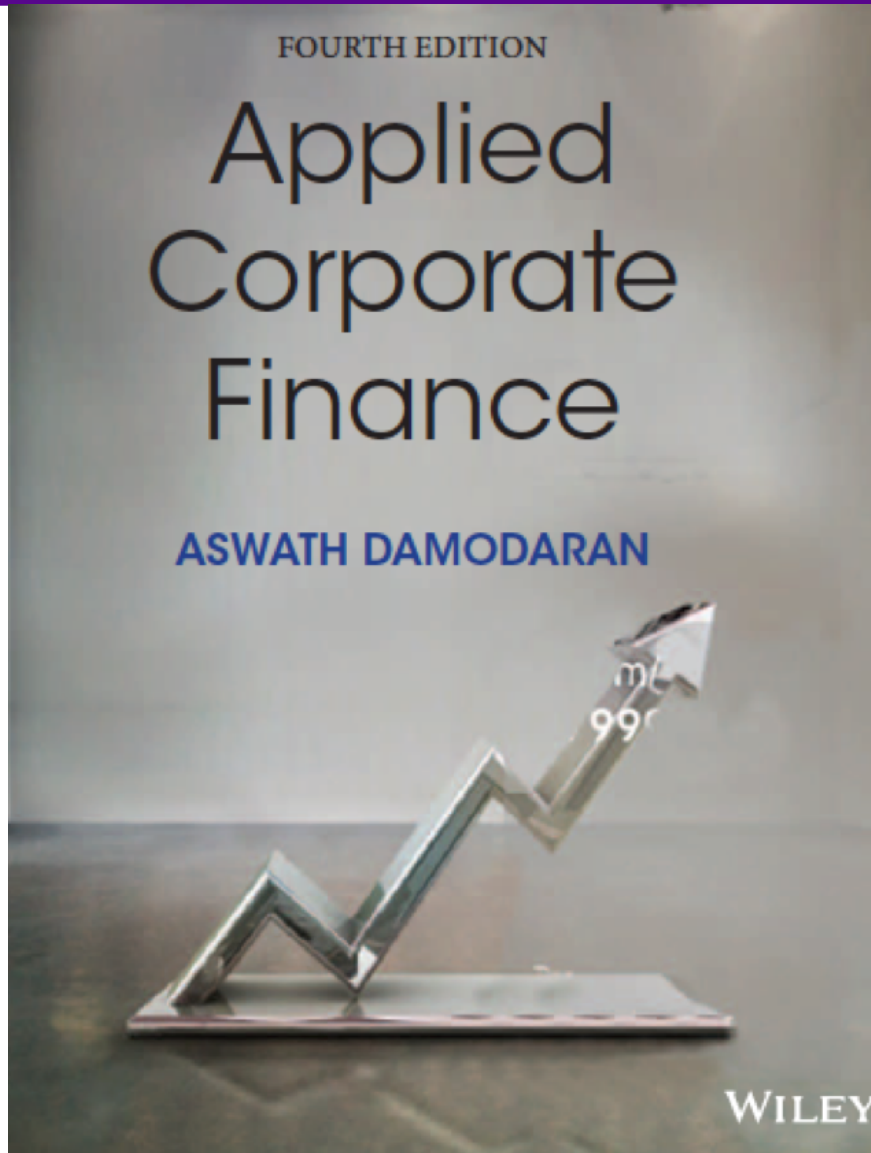




APPLICATION TEST: YOUR FIRM'S OPTIMAL FINANCING MIX

- Using the optimal capital structure spreadsheet provided,:
 - Estimate the optimal debt ratio of your firm
 - Estimate the new cost of capital at the optimal
 - Estimate the effect of the change in the cost of capital on firm value
 - Estimate the effect on the stock price
- In terms of the mechanics, what would you need to do to get to the optimal mix immediately?

Task
Examine the connection between your company's fundamentals and its optimal debt ratio



Optional:
Read
Chapter 8